



A Large Vision Language Model (LVLM)-Based System for Detecting Image-Text Misinformation

School of Computer Science & Applied Mathematics
University of the Witwatersrand

Tebogo Seopa
Student Number: 2563912

Supervised by Dr Seun Olukanmi

September 11, 2026

Abstract

Multimodal misinformation, where false or misleading content combines images and text to appear more convincing, has become one of the most serious challenges in online information ecosystems. Existing detection methods struggle because they were mostly designed for text-only content and fail to reason across both modalities at once. This project proposes a Large Vision Language Model (LVLM)-based system for detecting image-text misinformation that combines two-stage news-domain instruction tuning with targeted, query-driven external evidence retrieval. The system will be trained and evaluated on the COSMOS dataset, a well-established out-of-context image-caption benchmark, and tested for generalisation on MMFakeBench, which covers mixed-source forgery conditions. Performance will be measured using F1 score, precision, recall, and accuracy, alongside explanation quality assessment to evaluate the reasoning produced by the model. The research is situated in the field of artificial intelligence, specifically at the intersection of natural language processing and computer vision. The expected contribution is a detection system that is more accurate, explainable, and robust than existing approaches, along with a principled evaluation framework that goes beyond binary accuracy to assess the quality of the model’s reasoning.

Declaration

I, Tebogo Seopa, hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of Bachelor of Science with Honours in Big Data Analytics at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Acknowledgements

I would like to thank my supervisor, Dr Seun Olukanmi for her guidance and support throughout the development of this proposal. I also wish to acknowledge the researchers whose published work forms the foundation of this project.

Contents

Preface

Abstract	i
Declaration	ii
Acknowledgements	iii
Table of Contents	iv

1 Introduction 1

1.1 Background	1
1.2 Research Problem	2
1.3 Research Question	2
1.4 Aim	2
1.5 Objectives	2
1.6 Proposed Contribution	3
1.7 Overview of the Proposal	4

2 Related Work 5

2.1 The Problem of Misinformation at Scale	5
2.2 Pre-LLM Approaches to Multimodal Detection	5
2.3 LVLM-Based Detection: State of the Art	6
2.3.1 SNIFFER	6
2.3.2 LEMMA	6
2.3.3 FKA-Owl	6
2.3.4 Evidence Retrieval and Re-Ranking	7
2.3.5 End-to-End Fact-Checking with Explanations	7
2.3.6 MMFakeBench	7
2.4 Dataset Quality and Evaluation	7
2.5 Research Gaps	7

3 Research Methodology 9

3.1 Research Field and Approach	9
3.2 Dataset	9
3.2.1 Primary Dataset: COSMOS	9
3.2.2 Secondary Dataset: MMFakeBench	10
3.3 Modality Focus	10
3.4 Proposed System Architecture	11

3.4.1	Component 1: Two-Stage Instruction Tuning	11
3.4.2	Component 2: Targeted Query-Driven Evidence Retrieval with Re-Ranking	11
3.4.3	Component 3: Explanation Generation	12
3.5	Evaluation Framework	12
3.6	Tools and Implementation	13
4	Schedule of Work	14
5	Conclusion	16
	References	16

Chapter 1

Introduction

1.1 Background

We live in a time when a single social media post can reach millions of people in a matter of minutes, long before any journalist or fact-checker has had a chance to look at it. That speed has been useful for spreading important news quickly, but it has also created a serious problem: false and misleading content travels just as fast as the truth, and sometimes faster. The challenge is made significantly worse when that false content does not just consist of words but combines text and images together. People tend to trust content that looks visual and concrete. A real photograph placed next to a misleading caption can make a false claim feel entirely legitimate, even when the image has nothing to do with what the text says.

This kind of deception, where different types of media are combined to create or reinforce a false narrative, is what researchers call multimodal misinformation. Detecting it automatically has become one of the most urgent and difficult problems sitting at the crossroads of natural language processing and computer vision. The arrival of Large Language Models (LLMs) and Large Vision Language Models (LVLMs) has brought both new possibilities and new risks. On one side, these models are impressive reasoners with broad knowledge about the world, making them strong candidates for automated detection systems. On the other side, exactly the same capabilities mean they can be used deliberately to generate convincing false content at a scale and speed that was never possible before.

This proposal concerns the development of an LVLM-based system for detecting multimodal misinformation in image-text pairs. The research sits within the broader field of artificial intelligence, and more specifically within natural language processing and computer vision, two subfields that have traditionally operated somewhat separately but are increasingly being brought together through multimodal models. The project is not a medical, legal, or humanities project, though the problem it addresses has direct and serious consequences for public health, democratic processes, and society more broadly. The work is firmly technical and experimental in nature.

1.2 Research Problem

The core problem is that existing multimodal misinformation detection systems have three overlapping weaknesses that leave them poorly suited for real-world deployment. First, LVLMs reason poorly without external grounding. Several studies have shown that even strong models struggle to evaluate whether an image and caption are mismatched when no outside evidence is available COSMOS [2] and LEMMA [15], suggesting that internal model knowledge alone is not sufficient for reliable detection. Second, existing benchmarks are too simple. Most detection systems have been designed and evaluated on datasets that assume each piece of misinformation has a single forgery source, either a false caption or a manipulated image, but not both at once. Real-world misinformation is messier than that, and Liu et al. [7] demonstrate through MMFakeBench that current methods show significantly reduced performance under more realistic mixed-source conditions, with some models dropping as low as 25.7% F1. Third, explainability is largely absent. With very few exceptions such as SNIFFER [11], existing detectors produce a binary verdict but no explanation of why the content was flagged. That makes them difficult to trust and practically useless for journalists or fact-checkers who need to understand and communicate the reasoning behind a decision.

1.3 Research Question

This project is organised around the following research question:

Can a multimodal misinformation detection system built on domain-adapted LVLMs with targeted retrieval augmentation achieve better accuracy, better generalisation to mixed-source conditions, and more reliable explanations than existing approaches?

1.4 Aim

The aim of this project is to develop and evaluate an LVLM-based multimodal misinformation detection system that is more accurate, more explainable, and more robust to mixed-source conditions than existing approaches, by combining news-domain instruction tuning with targeted retrieval augmentation on the COSMOS dataset.

1.5 Objectives

The following objectives operationalise this aim:

1. **Systematic literature review and gap analysis.** Review the existing literature on multimodal misinformation detection with a focus on LVLM-based approaches, identify the key limitations and gaps in current methods, and establish clear benchmarks for comparison.
2. **System design and implementation.** Design and implement an LVLM-based detection pipeline that integrates: (a) two-stage instruction tuning for news-domain

alignment following the approach in SNIFFER [11]; (b) targeted query-driven evidence retrieval with LVLM-based re-ranking following LEMMA [15] and Tahmasebi et al. [13]; and (c) an explanation generation component that produces human-readable justifications alongside each verdict.

3. **Dataset preparation and experimental setup.** Prepare the COSMOS [2] dataset: the unlabelled training and validation splits (image-caption pairs with no out-of-context annotation) for Stage 1 entity-alignment tuning, and a stratified 70/15/15 split of the labelled test split – the only portion of COSMOS carrying ground-truth out-of-context labels – into training, validation, and held-out test sets for Stage 2 fine-tuning and final evaluation. Prepare MMFakeBench [7] for secondary evaluation to test generalisation under mixed-source conditions.
4. **Training and tuning.** Train Stage 1 (news-domain entity alignment) on COSMOS’s unlabelled training images and Stage 2 (out-of-context fine-tuning) on the labelled training split described above, tuning hyperparameters on the corresponding validation splits. Compare performance of different configurations, including with and without the retrieval component, and with and without the instruction tuning stage.
5. **Evaluation.** Evaluate the trained system on COSMOS’s held-out labelled test split and on MMFakeBench using F1 score, accuracy, and explanation quality metrics. Because SNIFFER and LEMMA report their published results on NewsCLIPPings rather than COSMOS, they are not used as a directly controlled comparison; instead, the system is compared against its own zero-shot LVLM baseline, reproduced on the same COSMOS test split, and against the ablation configurations described in Section 3.5, with SNIFFER, LEMMA, and COSMOS’s own reported figures cited as contextual reference points for the field’s general performance range.
6. **Analysis and reporting.** Analyse the results to understand where the system succeeds and fails, identify remaining limitations, and write up the findings in a full research report.

1.6 Proposed Contribution

The proposed system makes three contributions relative to the current state of the art. First, it combines two-stage instruction tuning for news-domain alignment with targeted query-driven evidence retrieval in a single unified framework, a combination that has not been explored in the literature. Second, it is explicitly designed to be evaluated under both single-source and mixed-source conditions, addressing the evaluation gap identified by Liu et al. [7]. Third, it treats explainability as a first-class goal rather than an afterthought, producing human-readable reasoning alongside each verdict and evaluating the quality of that reasoning alongside classification accuracy.

1.7 Overview of the Proposal

The rest of this proposal is structured as follows. Chapter 2 reviews the background literature and motivates the research problem in relation to existing work. Chapter 3 states the aims and objectives of the project. Chapter 4 describes the research methodology in detail. Chapter 5 presents the schedule of work. Chapter 6 concludes the proposal and summarises its expected contribution.

Chapter 2

Related Work

2.1 The Problem of Misinformation at Scale

The scale and speed at which misinformation spreads online has made manual detection entirely impractical. Patra et al. [10] survey fake news detection methods on social media and document the progression of the field from early rule-based systems through to machine learning and deep learning approaches. Their survey shows that the research community spent most of the 2010s focused on text-only content, using features like writing style, linguistic complexity, and social propagation patterns to classify claims as true or false. These methods worked reasonably well in controlled settings but were not designed for a world where fake news routinely involves images alongside text.

Chen and Shu [3] provide the most important recent framing of the problem. Their survey, published in AI Magazine, makes a point that is central to the motivation for this project: LLMs are a double-edged sword. The capabilities that make them useful for detection, including world knowledge, strong reasoning, and fluent language generation, are the exact same capabilities that make them effective at generating convincing misinformation. Their empirical finding is particularly sobering: misinformation produced by LLMs is harder to detect than human-written misinformation with the same meaning, both for human readers and for automated systems. This means the challenge is not static but is actively getting harder as generation capabilities improve.

Ai et al. [1] reinforce this in a recent comprehensive survey, documenting the shift from traditional feature engineering to end-to-end multimodal reasoning and identifying fine-grained semantic alignment as a persistent weakness across all current paradigms, particularly for detecting inconsistencies between text and image when neither looks obviously fake on its own.

2.2 Pre-LLM Approaches to Multimodal Detection

Before LLMs became available, researchers tackled the cross-modal detection problem using self-supervised learning and alignment techniques. Aneja et al. [2] propose COS-MOS, which detects out-of-context image-text pairs by grounding image content against

textual claims at the object level without explicit supervision. Their central insight is that language-only models are fundamentally insufficient for this task: a model that processes only text cannot determine whether two captions describe different visual content or simply use different words. COSMOS achieves 85% out-of-context detection accuracy on a 200,000-image dataset and establishes the importance of cross-modal grounding as the critical signal for this type of detection.

Sun et al. [12] advance this further with KDCN, a Knowledge-guided Dual-consistency Network that runs two simultaneous checks: one for visual-textual semantic mismatch and one for contradiction between post content and background knowledge from a knowledge graph. Tested on Twitter, Weibo, and PHEME, KDCN outperforms previous baselines and handles missing images gracefully using a special token mechanism. The conceptual insight that inconsistency between modalities is a reliable indicator of deception directly motivates the detection approach in this project.

2.3 LVLM-Based Detection: State of the Art

2.3.1 SNIFFER

Qi et al. [11] introduce SNIFFER, a two-stage instruction-tuned MLLM for out-of-context misinformation detection. The first stage aligns the model with news-domain entities, training it to identify named people and places rather than using generic descriptions. The second stage fine-tunes on out-of-context examples to teach the logic of cross-modal inconsistency detection. SNIFFER uses both internal image-text consistency checking and external evidence retrieval, combining both into a final verdict with a human-readable explanation. It outperforms the base MLLM by over 40% on NewsCLIPpings [8] and is one of the first systems to treat explainability as a core design goal. The instruction tuning approach and the internal-plus-external checking mechanism are both directly adopted in the proposed system.

2.3.2 LEMMA

Xuan et al. [15] demonstrate empirically that LVLMs consistently underperform when asked to verify claims without external evidence, even in zero-shot settings. LEMMA addresses this with a targeted retrieval pipeline where the model generates specific queries based on what it needs to know before retrieving evidence, keeping retrieved content focused rather than noisy. LEMMA improves accuracy by 7% on Twitter and 13% on Fakeddit [9] without any fine-tuning, making it portable across models and domains. The query-generation mechanism from LEMMA is incorporated into the proposed system.

2.3.3 FKA-Owl

Liu et al. [5] propose FKA-Owl, which augments LVLMs with two types of forgery-specific knowledge: semantic correlation between text and images, and artefact traces from image manipulation. A multi-level cross-modal reasoning module extracts semantic correlations while a dual-branch verification module captures spatial manipulation

traces. Experiments show FKA-Owl achieves better cross-domain generalisation than previous methods, a property particularly relevant to this project given that real-world misinformation does not respect domain boundaries.

2.3.4 Evidence Retrieval and Re-Ranking

Tahmasebi et al. [13] demonstrate the practical importance of evidence quality in detection pipelines. Their system, LVLM4FV, incorporates LVLM-based re-ranking to prioritise the most relevant image and text evidence before making a veracity prediction. This outperforms supervised baselines and generalises well across datasets, adding the contribution that re-ranking quality directly affects downstream detection accuracy.

2.3.5 End-to-End Fact-Checking with Explanations

Yao et al. [16] propose MOCHEG, a dataset of 15,601 claims each annotated with a truthfulness label, a ruling statement, and associated image and text evidence. Their end-to-end fact-checking system must retrieve relevant evidence, predict a verdict, and generate a natural language explanation. Their finding that state-of-the-art systems perform poorly when explanation quality is assessed alongside classification accuracy directly motivates this project’s treatment of explainability as a required evaluation criterion.

2.3.6 MMFakeBench

Liu et al. [7] introduce the first benchmark designed for mixed-source multimodal misinformation, covering three forgery types across twelve sub-categories: textual veracity distortion, visual veracity distortion, and cross-modal inconsistency. The benchmark contains 11,000 image-text pairs. When 15 LVLMs are evaluated under zero-shot conditions, all fail badly, with some dropping as low as 25.7% F1. This result is the clearest evidence available that the problem this project addresses is genuinely unsolved.

2.4 Dataset Quality and Evaluation

Thibault et al. [14] audit 75 misinformation datasets and find that most have serious quality problems, including spurious correlations and examples that cannot be verified for veracity even by human experts. They argue that standard metrics like accuracy and F1 cannot distinguish correct reasoning from lucky pattern-matching and propose the Evaluation Quality Assurance (EQA) framework as a more reliable evaluation standard. This directly informs the dataset selection and evaluation methodology of this project.

2.5 Research Gaps

Drawing together the reviewed literature, four specific gaps directly motivate this project.

Gap 1: Instruction tuning and targeted retrieval have not been combined. SNIFFER uses instruction tuning with internal and external checking. LEMMA uses targeted retrieval without fine-tuning. Neither paper explores what happens when the two

are combined: whether instruction tuning improves the quality of the query generation that drives retrieval.

Gap 2: Mixed-source detection is unsolved. All systems reviewed were designed for single-source conditions. MMFakeBench [7] shows all current methods fail under realistic mixed-source conditions.

Gap 3: Evaluation is inadequate. Binary metrics cannot distinguish correct reasoning from pattern-matching on dataset artefacts [14]. Yao et al. [16] show explanation quality must be evaluated separately from classification accuracy.

Gap 4: Explainability is not a primary design goal. Only SNIFFER [11] treats explainability as a core contribution. All other systems focus on classification accuracy alone.

Chapter 3

Research Methodology

3.1 Research Field and Approach

This project is an experimental computer science project. It sits at the intersection of two subfields of artificial intelligence: natural language processing (NLP) and computer vision (CV). More specifically, it belongs to the emerging area of multimodal AI, where models are designed to process and reason across more than one type of input simultaneously.

The approach is empirical and systems-based. The project proposes a detection system, implements it, trains it on real data, evaluates it against existing baselines, and draws conclusions from the results. There is no data collection from human subjects, no surveys, and no clinical or experimental protocols involving people. Ethics clearance is being sought for the use of publicly available datasets.

3.2 Dataset

3.2.1 Primary Dataset: COSMOS

The primary training and evaluation dataset for this project is COSMOS [2], introduced by Aneja et al. at AAAI 2023. COSMOS was originally created to support a self-supervised out-of-context detection method, discussed as related work in Section 2.2; this project uses only the dataset it introduced as a data source for the proposed LVLM-based system, not the original self-supervised method.

The dataset consists of images drawn from a wide range of news and fact-checking websites, each associated with one or more captions scraped from the articles they originally appeared in. It is split into a training split and a validation split, together covering over 160,000 images, and a test split of approximately 1,700 images. Critically, only the test split carries a ground-truth out-of-context label; the training and validation splits are provided as raw image-caption associations with no annotation, since COSMOS’s original method is self-supervised and does not require labelled training data. The test split also has a different structure from a single image-caption pair: each test image is paired with two candidate captions, and the label indicates whether pairing both captions

with that image constitutes an out-of-context use, rather than describing either caption individually as true or false.

This project uses both parts of the dataset for different purposes. The large unlabelled training and validation splits are used for Stage 1 news-domain entity-alignment tuning (Section 3.4.1), which requires only genuine image-caption pairs and no out-of-context labels. The small labelled test split, further divided into a stratified 70/15/15 train/validation/test split, is used for Stage 2 out-of-context fine-tuning (Section 3.4.1) and for final evaluation.

Change of primary dataset. This project originally proposed NewsCLIPpings [8] as the primary dataset. That decision was revised, with the supervisor’s approval, after the NewsCLIPpings image corpus (VisualNews, distributed as a single 91GB archive) proved impossible to download intact despite repeated attempts over several days, including two separate file-corruption incidents traced to the hosting server’s connection. COSMOS was selected as the replacement because it addresses the same out-of-context detection task, is already discussed in the literature review (Section 2.2), and is distributed as a small number of independently hosted files rather than one large archive.

This change has one direct methodological consequence, addressed explicitly in the evaluation framework (Section 3.5): because only around 1,700 COSMOS images are labelled, the labelled data available for Stage 2 fine-tuning and evaluation – roughly 1,190 training, 255 validation, and 255 test images after the 70/15/15 split – is substantially smaller than the 35,536 training examples NewsCLIPpings would have provided. Stage 1 is unaffected, since it uses the large unlabelled portion of COSMOS and does not depend on out-of-context ground truth.

3.2.2 Secondary Dataset: MMFakeBench

MMFakeBench [7] will be used for secondary evaluation to test whether the trained system generalises beyond the single-source assumptions of COSMOS. It contains 11,000 image-text pairs across three forgery sources and twelve sub-categories, making it a significantly harder and more realistic benchmark. No training will be done on MMFakeBench; it will be used for evaluation only. This follows the convention established by SNIFFER [11] and is consistent with the goal of testing generalisation.

3.3 Modality Focus

This project focuses on the image-text modality combination. This choice is deliberate. Image-text misinformation, particularly out-of-context pairs where a real photograph is paired with a misleading caption, is the dominant form of multimodal deception on mainstream social media platforms. Its effectiveness comes from the fact that both components look individually legitimate: there is no obvious artefact in the image and no clearly false language in the text. Detecting it requires genuine cross-modal reasoning, which is exactly the capability this project aims to develop and evaluate.

Video and audio introduce substantially greater complexity in terms of temporal reasoning, audio-visual synchrony, and computational requirements. Restricting the scope

to image-text allows this project to go deeper on a well-defined, practically important problem rather than spreading too thinly across multiple modalities. This focus is also consistent with all the most relevant existing systems, SNIFFER [11], LEMMA [15], FKA-Owl [5], and MMFakeBench [7], all of which work with image-text pairs, enabling direct comparison.

3.4 Proposed System Architecture

The proposed system has three main components, each motivated by a specific gap in the existing literature.

3.4.1 Component 1: Two-Stage Instruction Tuning

The base model will be LLaVA [6], an open-source LVLM with strong visual-language alignment and an active research community around it. It will be adapted through a two-stage instruction tuning process following the approach established by SNIFFER [11].

Stage 1 – News-domain entity alignment. General-purpose LVLMs describe images using broad category terms. When shown a photograph of a politician, for example, they will say “a man in a suit” rather than identifying the specific person. This is a serious limitation for news-domain misinformation detection, where the identity of the person in the photograph is often precisely what matters. The first tuning stage trains the model on image-caption pairs from news sources to align its visual representations with specific named entities, teaching it to describe images in terms of the people, places, and objects that are actually relevant to fact-checking. This stage closes the domain gap between a general-purpose LVLM and a news-domain detection system.

Stage 2 – Out-of-context detection fine-tuning. The second stage teaches the model the particular logic of out-of-context detection using instruction-following examples. Each example presents an image-text pair and asks the model to assess whether the image and text match, and to explain its reasoning. This teaches the model not just what out-of-context misinformation is, but how to reason about whether a specific pair is an instance of it.

3.4.2 Component 2: Targeted Query-Driven Evidence Retrieval with Re-Ranking

Following the approach from LEMMA [15] and Tahmasebi et al. [13], the system will incorporate an external evidence retrieval step that runs after the initial domain-aligned assessment and before the final verdict is produced.

The model first generates a set of targeted retrieval queries based on what it has identified as the key factual claims in the image-text pair. This is more effective than issuing a single broad query because it directs retrieval toward the specific information needed to verify or falsify the claim. Retrieved evidence is then re-ranked using the LVLM itself

to filter out irrelevant results and prioritise the most probative evidence for the final decision.

The retrieved evidence, consisting of related images and textual documents from web sources, is then incorporated into a final verification step where the model reasons across the original image-text pair and the retrieved evidence together to produce a verdict.

This component addresses Gap 1 identified in the literature review by combining targeted retrieval with task-specific instruction tuning in a single system. Whether instruction tuning improves the quality of the query generation step, and whether re-ranking further improves detection accuracy, are both testable experimental questions.

3.4.3 Component 3: Explanation Generation

Every verdict produced by the system will be accompanied by a human-readable explanation of the reasoning behind it. The explanation will identify the specific evidence that the system found most probative, describe the inconsistency or consistency detected between the image and the text, and state the conclusion with a confidence indication.

This component addresses Gap 4 from the literature review. It is also practically important: a detection system that simply outputs “fake” or “real” without reasoning is of limited use to journalists, fact-checkers, or platform moderators who need to understand and communicate why a particular piece of content is problematic.

3.5 Evaluation Framework

The evaluation framework is designed to go beyond binary accuracy and F1, consistent with the critique by Thibault et al. [14] and the precedent set by Yao et al. [16].

Classification metrics. Standard F1 score (macro-averaged), precision, recall, and accuracy on COSMOS’s held-out labelled test split and on MMFakeBench.

Generalisation. Performance on MMFakeBench without any training on that dataset, to test how well the system handles the more complex mixed-source conditions it was not explicitly trained for.

Explanation quality. A sample of the system’s generated explanations will be evaluated for factual accuracy, coherence, and relevance. This will be done both using automated metrics and, where feasible, through a small human evaluation study.

Ablation study. The system will be evaluated in several configurations: without instruction tuning, without the retrieval component, without re-ranking, and without explanation generation, all on the same held-out COSMOS test split. Given the small size of COSMOS’s labelled data (Section 3.2), the ablation study is treated as the primary evidence for each component’s contribution, since every configuration is compared on identical, held-out data with no cross-dataset confound.

Baseline comparison. SNIFFER [11] and LEMMA [15] report their published results on NewsCLIPpings rather than COSMOS, so their figures cannot be used as a directly

controlled, same-dataset comparison. Results will instead be compared against: (1) a zero-shot LVLMM baseline with no instruction tuning or retrieval, reproduced on the same COSMOS test split as every other configuration; and (2) SNIFFER, LEMMA, and COSMOS’s own reported self-supervised results [2], cited as contextual reference points for the general performance range achieved in this field rather than as a controlled comparison.

3.6 Tools and Implementation

The project will be implemented in Python using the Hugging Face Transformers library for model loading and fine-tuning. LLaVA will be loaded from the official Hugging Face repository. Fine-tuning will use parameter-efficient methods such as LoRA (Low-Rank Adaptation) to make the training feasible on available hardware. Evidence retrieval will use a combination of web search APIs and local retrieval from a pre-indexed evidence corpus. Experiments will be tracked using Weights and Biases. All code will be version-controlled using Git and made available in a project repository.

Chapter 4

Schedule of Work

The project will run from May 2026 through to submission in October 2026. The schedule is divided into six phases. Table 4.1 summarises the planned timeline.

Table 4.1: Project schedule of work

Phase	Period	Activities
Phase 1: Setup and literature	May 2026	Final literature review; environment setup; dataset acquisition and preprocessing
Phase 2: Baseline experiments	June 2026	Implement zero-shot LVLM baseline on COSMOS; establish evaluation framework; contextualise against SNIFFER and LEMMA’s published figures
Phase 3: Stage 1 tuning	July 2026 (weeks 1–2)	Implement and run news-domain entity alignment tuning; validate alignment against baseline
Phase 4: Full system	July 2026 (weeks 3–4) – August 2026	Stage 2 fine-tuning; retrieval pipeline implementation; re-ranking; explanation generation; full system integration
Phase 5: Evaluation	September 2026	Full evaluation on COSMOS and MM-FakeBench; ablation studies; explanation quality assessment
Phase 6: Write-up	October 2026	Analysis of results; writing full research report; final submission

Each phase has a clear deliverable: Phase 1 produces a finalised proposal and clean datasets; Phase 2 produces baseline results; Phases 3 and 4 produce the trained system; Phase 5 produces all evaluation results; Phase 6 produces the final report. The schedule

is designed with one week of buffer in each of Phases 4 and 5 to accommodate unexpected technical difficulties with training or evaluation.

Chapter 5

Conclusion

This proposal has described a research project to develop an LVLM-based multimodal misinformation detection system that combines news-domain instruction tuning, targeted query-driven evidence retrieval with re-ranking, and explicit explainability in a unified architecture. The project is motivated by four clear gaps in the existing literature: the combination of instruction tuning and targeted retrieval has not been explored; mixed-source detection remains effectively unsolved; standard evaluation metrics are inadequate for LLM-based detection systems; and explainability is not treated as a primary design goal in most current systems.

The system will be trained and primarily evaluated on the COSMOS dataset, chosen after the originally proposed NewsCLIPpings image corpus proved impossible to acquire reliably, for its direct alignment with the cross-modal reasoning capability the project aims to build and its established position in the literature reviewed in Chapter 2. It will be tested for generalisation on MMFakeBench, the most realistic benchmark currently available. The evaluation framework goes beyond binary accuracy to include explanation quality assessment and a thorough ablation study.

The expected contribution of this project is threefold. First, it will produce a detection system that outperforms existing approaches on both single-source and mixed-source evaluation conditions. Second, it will demonstrate the value of combining instruction tuning with targeted retrieval in a principled way, filling a gap in the literature that currently separates these two approaches. Third, it will produce explanations alongside verdicts, making the system more useful in practice and more honest about its own reasoning process. Together, these contributions represent a meaningful step forward in the development of reliable, explainable, and practically deployable multimodal misinformation detection systems.

References

- [1] W. Ai, Y. Tan, Y. Shou, T. Meng, et al. The Paradigm Shift: A Comprehensive Survey on Large Vision Language Models for Multimodal Fake News Detection. *arXiv preprint arXiv:2601.15316*, 2025. <https://arxiv.org/abs/2601.15316>
- [2] S. Aneja, C. Bregler, and M. Nießner. COSMOS: Catching Out-of-Context Misinformation with Self-Supervised Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, pages 14084–14092, 2023. <https://arxiv.org/abs/2101.06278>
- [3] C. Chen and K. Shu. Combating Misinformation in the Age of LLMs: Opportunities and Challenges. *AI Magazine*, 45(3):354–368, 2024. <https://doi.org/10.1002/aaai.12188>
- [4] Complex Data Lab. CDL Misinfo Datasets: A Repository for Misinformation Datasets and Benchmarks. Online resource, 2024. <https://misinfo-datasets.complexdatalab.com/>
- [5] X. Liu, P. Li, H. Huang, Z. Li, X. Cui, J. Liang, L. Qin, W. Deng, and Z. He. FKA-Owl: Advancing Multimodal Fake News Detection through Knowledge-Augmented LVLMS. In *Proceedings of the 32nd ACM International Conference on Multimedia (ACM MM)*, pages 10154–10163, 2024. <https://doi.org/10.1145/3664647.3681089>
- [6] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual Instruction Tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 34892–34916, 2023. <https://arxiv.org/abs/2304.08485>
- [7] X. Liu, Z. Li, P. Li, H. Huang, S. Xia, X. Cui, L. Huang, W. Deng, and Z. He. MM-FakeBench: A Mixed-Source Multimodal Misinformation Detection Benchmark for LVLMS. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. <https://arxiv.org/abs/2406.08772>
- [8] G. Luo, T. Darrell, and A. Rohrbach. NewsCLIPPings: Automatic Generation of Out-of-Context Multimodal Media. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6801–6817, 2021. <https://doi.org/10.18653/v1/2021.emnlp-main.545>

- [9] K. Nakamura, S. Levy, and W. Y. Wang. r/Fakeddit: A New Multimodal Benchmark Dataset for Fine-Grained Fake News Detection. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, pages 6149–6157, 2020.
- [10] D. Patra, B. Jana, S. Mandal, and A. A. Sekh. Understanding Fake News Detection on Social Media: A Survey on Methodologies and Datasets. In A. A. Sekh et al. (Eds.), *Artificial Intelligence: ISAI 2022. Communications in Computer and Information Science*, vol. 1695, pages 243–255. Springer, 2022. https://doi.org/10.1007/978-3-031-22485-0_21
- [11] P. Qi, Z. Yan, W. Hsu, and M. L. Lee. SNIFFER: Multimodal Large Language Model for Explainable Out-of-Context Misinformation Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13052–13062, 2024. <https://doi.org/10.1109/CVPR52733.2024.01240>
- [12] M. Sun, X. Zhang, J. Ma, S. Xie, Y. Liu, and P. S. Yu. Inconsistent Matters: A Knowledge-Guided Dual-Consistency Network for Multi-Modal Rumor Detection. *IEEE Transactions on Knowledge and Data Engineering*, 35(12):12736–12749, 2023. <https://doi.org/10.1109/TKDE.2023.3275586>
- [13] S. Tahmasebi, E. Müller-Budack, and R. Ewerth. Multimodal Misinformation Detection using Large Vision-Language Models. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM)*, pages 2189–2199, 2024. <https://doi.org/10.1145/3627673.3679826>
- [14] C. Thibault, J.-J. Tian, G. Peloquin-Skulski, T. L. Curtis, J. Zhou, F. Laflamme, Y. Guan, R. Rabbany, J.-F. Godbout, and K. Pelrine. A Guide to Misinformation Detection Data and Evaluation. *arXiv preprint arXiv:2411.05060*, 2024. <https://arxiv.org/abs/2411.05060>
- [15] K. Xuan, L. Yi, F. Yang, R. Wu, Y. R. Fung, and H. Ji. LEMMA: Towards LVLm-Enhanced Multimodal Misinformation Detection with External Knowledge Augmentation. *arXiv preprint arXiv:2402.11943*, 2024. <https://arxiv.org/abs/2402.11943>
- [16] B. M. Yao, A. Shah, L. Sun, J.-H. Cho, and L. Huang. End-to-End Multimodal Fact-Checking and Explanation Generation: A Challenging Dataset and Models. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 2733–2743, 2023. <https://doi.org/10.1145/3539618.3591879>
- [17] F. Zeng, W. Li, W. Gao, and Y. Pang. Multimodal Misinformation Detection by Learning from Synthetic Data with Multimodal LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 10467–10484, 2024. <https://doi.org/10.18653/v1/2024.findings-emnlp.613>